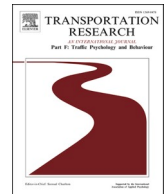




Contents lists available at ScienceDirect

Transportation Research Part F: Psychology and Behaviour

journal homepage: www.elsevier.com/locate/trf

In-vehicle human–machine interface guidelines for augmented reality head-up displays: A review, guideline formulation, and future research directions

Putra A.R. Yamin^{a,b}, Jaehyun Park^{a,c,*}, Hyun K. Kim^d^a Department of Industrial & Management Engineering, Incheon National University (INU), South Korea^b Department of Industrial Engineering, Bandung Institute of Technology, Indonesia^c Center for Perception and Behavioral Research, Incheon National University (INU), South Korea^d School of Information Convergence, Kwangwoon University, South Korea

ARTICLE INFO

Keywords:

Augmented reality head-up display
Intelligent vehicle systems
Human-machine interface guideline
Ergonomic design

ABSTRACT

Background: The augmented reality head-up display (AR-HUD) recently emerged as a state-of-the-art in-vehicle display. However, no comprehensive guidelines apply to the human–machine interface for AR-HUD.

Objective: We formulated guidelines for the human-vehicle interface to develop an AR-HUD.

Methods: A review of existing literature on AR-HUD and applicable human factors and human–computer interaction guidelines was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach. In total, 121 articles were selected, and an initial set of guidelines for developing an AR-HUD was drawn up. We analyzed AR-HUD development and categorized the articles based on HMI features relevant to in-vehicle AR-HUD: virtual image distance, field of view, and brightness. Furthermore, research findings on projected information and legibility were investigated.

Results: Nine guidelines were devised for developing the AR-HUD human–machine interface. This study also compared the traditional AR-HUD and windshield AR-HUD, defining the requirements for future studies using the term AR-HUD and providing a framework for AR-HUD design.

Conclusions: Existing research on AR-HUD is limited. Future research is needed on virtual image distance, field of view, brightness, and legibility. The findings of this study could be used to guide the design of a full-windshield AR-HUD interface and future research.

1. Introduction

Augmented-reality head-up display (AR-HUD) is an advanced in-vehicle display technology that aims to provide drivers with a safer and better driving experience (Gabbard et al., 2019). The AR-HUD technology has been around for a while, but recent developments have made it more practical, accessible, and efficient. In recent years, there has been a significant increase in the adoption of AR-HUD technology in cars due to the many benefits it offers. AR-HUD utilizes real-time sensor information and progressions in internet-of-things and autonomous systems, navigational prompts, and advanced driver assistance systems (ADAS) and then presents information into the driver's field of view (Abdi & Meddeb, 2018; Topliss et al., 2019). Therefore, it can increase attention to cued

* Corresponding author at: Department of Industrial & Management Engineering, Incheon National University (INU), South Korea.
E-mail address: jaehpark@inu.ac.kr (J. Park).

<https://doi.org/10.1016/j.trf.2024.06.001>

Received 24 November 2023; Received in revised form 3 June 2024; Accepted 3 June 2024

Available online 12 June 2024

1369-8478/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

elements without adversely affecting attentional resources and decreasing the ability to respond to environmental information outside the focus of attention (Abdi & Meddeb, 2018; Lisle et al., 2019).

While traditional AR-HUDs provide drivers with digital information overlaid onto their line of sight, new full-windshield AR-HUDs take this technology a step further by projecting information across the entire windshield. This means that drivers can see digital information not only in a small section of their field of view but also in peripheral vision, providing a more immersive and natural driving experience. Full windshield AR-HUDs often use advanced artificial intelligence and machine learning algorithms to analyze data from various sensors and cameras to provide drivers with real-time information about their surroundings, such as weather conditions, traffic, and potential hazards (Murugan et al., 2022). This technology has the potential to greatly enhance driver safety and convenience by providing a more complete and informative view of the driving environment.

One significant development in AR-HUD technology is the increased use of advanced sensors that detect environmental conditions, such as rain, snow, and fog, to enhance the driver's safety (Zhang et al., 2021). AR-HUD displays can now adapt to changing weather conditions and display the necessary information on the windshield. Secondly, AR-HUD technology can also display navigation information, such as turn-by-turn directions, lane guidance, and speed limits, helping drivers stay on track and avoid accidents. Thirdly, another development in AR-HUD technology is the use of advanced algorithms and machine learning to analyze and interpret data collected from various sensors (Zhan et al., 2020). Drivers are met with numerous elements requiring their attention while driving. The present research focuses on selected visual and cognitive distractions faced by the driver and their influence on detecting and perceiving changes in the traffic environment. The main findings show that thinking about personal problems, chores, and errands, as well as roadside advertisements on the cognitive side and looking at advertisements and the natural environment on the visual side, present the most negative impacts on drivers' perception of crucial changes in the traffic environment. On the other hand, drivers who visually focus on traffic signals and pedestrians and think about driving speed, driving rules, and other traffic participants tend to notice crucial changes in the traffic environment more often (Cvahte Ojsteršek & Topolšek, 2019). With these algorithms, AR-HUD displays can now provide drivers more accurate and personalized information. For instance, it can learn the driver's behavior and adjust the display accordingly. This means that the display can be customized to show only important information to aid the driver in a safe manner.

Finally, the development of AR-HUD technology has also seen the integration of various features, such as gesture recognition and voice commands (Zhou & Yin, 2021). With these features, drivers can interact with the display without taking their hands off the wheel or their eyes off the road. The gesture recognition feature allows drivers to control the display by simply waving their hands, while the voice command feature allows them to give commands and receive responses audibly. These features make AR-HUD displays more intuitive and easier to use, further enhancing the driving experience.

The AR-HUD is no longer just an option to showcase the technology and luxury of the car manufacturer. With the increase in vehicles and pedestrians and our complex lifestyle, drivers need to be aware of more objects and information on the road and respond to more unpredictable situations. AR-HUD has the potential to become common equipment in all types of vehicles and will effectively contribute to the safety of drivers and pedestrians.

To be defined as AR-HUD, unlike traditional HUDs, the augmented image should overlay real-world objects like other cars, pedestrians, traffic lights, etc. (Yontem et al., 2021). The purpose is to increase the driving experience and driver safety by providing images that make it easier for the driver to attend to lanes, traffic signs, and other vehicles or objects (Calvi et al., 2020b). However, displaying excessive information to the driver could result in information overload (Eyraud et al., 2015; Naujoks et al., 2015). AR applications might distract drivers, leading them to take their eyes off the road or engage them in other tasks (Kim & Gabbard, 2019; Valimont et al., 2007). To prevent distraction, an entire windshield AR-HUD interface design should be guided by human factors display design and human-computer interaction principles (Merenda et al., 2018; Park & Park, 2019). Park and Park (2019) comprehensively reviewed functional requirements for traditional automotive HUDs. However, to our best knowledge, no guidelines apply specifically for designing AR-HUD human-machine interfaces to designing AR-HUD human-machine interfaces (HMIs). This article is an extension of (Yamin et al., 2021), which was a preliminary study toward formulating guidelines for in-vehicle AR-HUD design. This article conveyed a systematic review from an updated list of references, which resulted in a more complete and rigorous review.

Briefly, the objectives of this study are the following: (1) To provide a systematic overview of existing research associated with the human-vehicle interface for AR-HUD. (2) Formulate guidelines for the human-vehicle interface to develop AR-HUD. (3) Differentiate traditional versions of AR-HUD with windshield AR-HUD. (4) Provide an AR-HUD design conceptual framework (5) Outline the key areas where future research on AR-HUD is needed.

A review of existing literature on AR-HUD was conducted to map out potential future experiments. The research methodology is provided in Section 2. Section 3 develops the results and discussion of the current research in AR-HUD HMI. Furthermore, section 4 provides the proposed HMI guidelines for AR-HUD. Subsequently, in section 5, a debate on AR-HUD terminology, conceptual framework, implications, applications, and future research is included. Finally, the conclusion is presented in section 6.

2. Method

This study conducted a comprehensive literature review following the revised PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) method (Liberati et al., 2009), a well-accepted methodology for conducting and reporting systematic reviews in any engineering domain. Articles published from 2004 to 2024 were reviewed. The online databases utilized were Web of Science Core Collection and Science Direct. The year 2004 was chosen to ensure the comprehensiveness, timeliness, and relevance of the results in making a new set of guidelines.

The search was conducted using a complete dual review approach (Stoll et al., 2019). Then, a third researcher reviewed the identification and screening process. This ensured all relevant studies were included in this review. A wide range of documents, including journal articles, conference papers, technical reports, and relevant guidelines, were included in the search scope. The search strings selected for this literature search were (automotive OR car OR driving) AND ((augmented AND reality) AND “head-up display”). Further exploration of topically related and interchangeable terms was conducted, shown in Table 1. The rationale for using these variants of the term “AR-HUD” is that these phrases are used interchangeably in the literature.

Papers whose abstract mention an analysis of augmented reality interface design for vehicles as part of the results are included for further review. Papers that contain information regarding the interface design requirement for AR devices were included. With the objective of deriving a comprehensive guideline for AR-HUD HMI, relevant existing guidelines and reports were included in this study. For example, the development of design and engineering recommendations for in-vehicle alphanumeric messages (SAE J2831_201204) was included in this review. The studies not written in English and articles not available for retrieval were excluded from this review. All selected articles' content was analyzed, and the independent variables, dependent variables (if applicable), and results were compiled.

After reviewing the literature, information related to HMI for AR-HUD could be categorized into the development of in-vehicle AR-HUD, virtual image distance (VID), field-of-view (FoV), brightness, projected information, and legibility. Some articles provided information on more than one category. Some articles only had one. For example, studies that developed and evaluated a certain technology or prototype regarding AR-HUD are categorized under the development of in-vehicle AR-HUD. VID refers to an AR-HUD system's apparent distance between the projected image and the driver's eyes. It is a critical factor in determining the visual comfort and effectiveness of the AR-HUD display. Field of view (FoV) is the extent of the environment visible to the driver through the AR-HUD system. It represents the angle of the display's projection that covers the driver's visual field—projected information regarding what and how to present in certain conditions. Projected information regards the type of AR information needed in certain situations. Legibility refers to the ease with which drivers can read and understand the information displayed on the screen while driving.

3. Systematic literature review

The process of identification via databases is presented in Fig. 1 below. As a result, 146 and 328 documents were obtained from Web of Science Core Collection and Science Direct, respectively, totaling 474 records identified from the search. After examining each title, 107 documents listed from the Web of Science database were found in Science Direct. Hence, these 107 duplicate records were removed. Papers whose abstract mention an analysis of augmented reality interface design as part of the results are included for further review. Papers that contain information regarding the interface design requirement for AR-HUD were included. A further 214 articles irrelevant to AR-HUD for automotive and 18 not available in English were excluded, resulting in 135 articles sought for retrieval. Ten articles were unavailable to be retrieved, and four more were excluded due to being written by the same authors but having very similar results to those of another published paper. These excluded articles were conference papers published before an extended version of the research in a journal paper. As a result, 121 critical articles relevant to this study were selected in this review. The number of articles based on its type was 74 journal papers, 40 conference proceedings, two technical reports, and five Human Factors and Human-Computer Interaction Guidelines.

Based on its content category, the articles are summarized in 3.1 to 3.4. Based on the available information on VID, FoV, brightness, projected information, and legibility, the selected articles and a summary of the variables used in the study are presented in Table A1 (Appendix).

3.1. In-vehicle HMI development for AR-HUDs

In recent years, significant developments have occurred in the field of AR-HUD, with advancements in technology driving the creation of new and innovative applications. In 2013, Kim et al. created a simulation of an aftermarket HUD that presented semi-transparent AR forward collision warning and blind spot warning via a virtual display with a $24 \times 8^\circ$ field of view (Kim et al., 2013). The study shows that head-up AR display interfaces for crash warning systems have potential safety benefits and a high likelihood of driver acceptance. Schall et al. also simulated an entire windshield HUD for AR collision warning in the same year, using virtual boxes integrated into the driving scene to highlight road hazards (Schall et al., 2014). The study concluded that AR cues promise to improve elderly driver safety by increasing hazard detection likelihood without interfering with other driving tasks, such as maintaining safe headway. Lauber and Butz (2013) simulated a head-up display using an LCD and combiner mirror with 70 % transparency, comparing the screen-fixed presentation of speed, speed limit, collision warning, and basic navigation information to the head-worn AR presentation. The results showed that when using head-stabilized visualizations, the content can be displayed to drivers when they are distracted or turning their heads away from the road. For safety, depending on which direction the user turns their head,

Table 1
Search keywords for literature search.

Concept	Search keywords
AR	Augmented reality, AR
HUD	Head-up display, Head-up Display, HUD
Vehicle	Car, automotive, automobile, vehicle, in-vehicle, driving

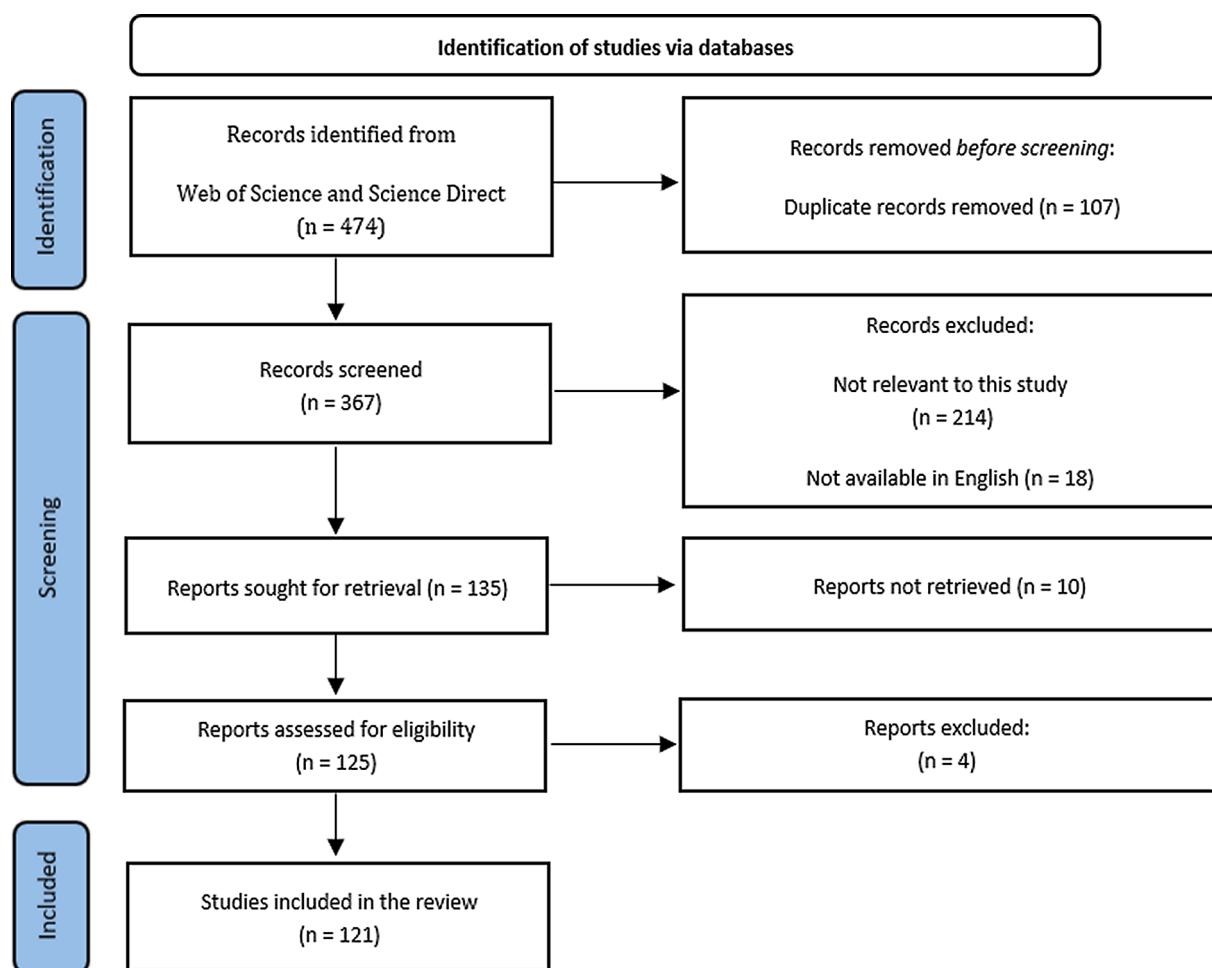


Fig. 1. Flow diagram of systematic review.

head-stabilized content could be dynamically re-arranged to prevent it from blocking the driver's vision.

In 2015, Pfannmüller et al. used a contact analog head-up display (cHUD) mock-up to explore various AR navigation display concepts by presenting AR graphics atop a video projection of driving scenes. This simulation allowed for a quick assessment of AR HMI design concepts but did not support manual driving or investigation of conformal AR graphics. Pauzie (2015) explored head-up displays' potential benefits and drawbacks in automotive applications. The study highlights the potential benefits of head-up displays, including improved safety, situational awareness, and driver comfort, and identifies potential drawbacks, such as distraction and increased complexity.

Smith et al. (2016) compared head-up and head-down displays using traditional methods of display assessment in a driving simulator. The head-up display provided better driving performance, situational awareness, and usability than the head-down display. This finding aligns with a study by Ma et al. (2021), which examined the effects of an AR-HUD on driving performance using a virtual reality simulator and eye-tracking technology. Sburlan et al. (2020) proposed an adaptive positioning HMI based on the driver's head orientation. Abu-Khadrah et al. (2023) designed an Adaptable Recommendation System (ARS) using an analytical CV (ACV) system for detecting objects using pervasive computing in intelligent transportation. The proposed method achieved better detection accuracy, less detection time and error, and improved precision compared to existing systems. Liu et al. (2023) evaluated windshield and sunroof AR-HUD prototypes for thermal and acoustic comfort by developing glass heated by coating, heat-reflecting glazing, and acoustic glazing based on marketing needs and users' demands.

3.2. In-vehicle AR-HUD HMI features

The article by Betancur et al. (2018) provides an overview of research topics and implementation trends related to automotive HUD systems. The authors note that HUDs have the potential to enhance driver safety and comfort by providing important information without the need for drivers to take their eyes off the road. The article discusses various types of HUDs, including combiner-type, windshield-type, and spatial augmented reality HUDs. It highlights the importance of considering factors such as display location,

the field of view, and brightness when designing HUD systems. The authors also review research on the effects of HUDs on driver performance, including studies on visual attention, reaction time, and driving behavior. They conclude that HUDs can potentially improve driver safety and comfort, but that further research is needed to optimize their design and evaluate their impact on driver performance in real-world driving conditions.

In an AR-HUD system, a transparent display panel projects virtual images of various driving-related information such as speed, navigation directions, and alerts onto the vehicle's windshield. The virtual image distance is between the driver's eyes and the projected virtual image, which appears floating in space. According to Li et al., 2019, virtual image distance should be carefully controlled to avoid visual discomfort or distortion of the virtual images. The authors suggest that the optimal virtual image distance in an AR-HUD system should be between 2 and 8 m, depending on the specific design of the display system and the driving environment. However, Texas Instruments reported that based on their study, the VID requirement is at least 6 m (Texas Instruments, 2021). The previous report by the same company (Firth, 2019) stated that 7 m is the minimum. However, it was revised to 6 m. According to Texas Instrument's 2021 report, the most suitable VID is 12 to 15 m. More than 15 m, any slight deviation of the augmented graphic from this range will result in less significant benefits beyond this distance.

Broy et al. (2014) explored the design parameters for a 3D HUD. They found that 3D head-up displays can provide better depth perception and improved safety, but design parameters must be optimized for optimal performance. Lisle et al. (2019) investigated the effects of volumetric augmented reality displays on human depth perception using a visual matching task. The study finds that volumetric augmented reality displays can negatively affect depth perception, particularly for tasks that require precise depth judgments. Different visual parameters, such as FoV and eye position, can affect the perception of perspective in augmented reality head-up displays (Bremers et al., 2021) and cognitive tunneling (Pullukat et al., 2020). Lv et al. (2021) presented a multi-plane augmented reality head-up display system that uses volume holographic optical elements with a large area. The proposed system can provide multiple depth planes and achieve a wide field of view. A new holographic waveguide AR-HUD that is optimized for volume and does not require mirrors was presented by Richter et al. (2018). The proposed display was found to provide high image quality, a wide field of view, and reduced system complexity compared to traditional head-up displays. User satisfaction was measured with this technology for traffic information applications (Abdi et al., 2015; Wang et al., 2018).

According to Gabbard et al. (2019), the integration of graphics directly into the driving scene through computer graphics or video does not provide the same accommodative and cognitive switching capabilities as a separate AR display. This is an essential consideration for research that aims to accurately investigate AR-HUDs' effects on a driver's visual attention. In addition, homemade AR-HUDs, such as those created using tablets and semi-transparent combiners, may be susceptible to visual artifacts like ghosting that can impact the results of user studies unless the construction is executed with extreme care. Another issue that needs to be noted is frequent context and focal distance switching can impair driver performance (Gabbard et al., 2019).

Ren et al. (2016) evaluated the effectiveness of wide-FoV augmented reality displays in a mixed-reality simulation of driving scenarios. The wide-FoV AR display was found to improve driving performance and decrease response time to critical events. They suggested a wide field of view of at least 10° is needed to present pedestrians and buildings. A wider FoV in AR-HUD can enhance driver awareness and safety by providing more information about the driving environment (Kim, Isleib, et al., 2016; Kim, Miranda Anon, et al., 2016). However, an excessively wide FoV can also result in visual clutter and distraction for the driver. Therefore, the FoV should be carefully designed and calibrated to balance the amount of information presented to the driver and the clarity of the display (Gao et al., 2019).

In conclusion, the optimal FoV for AR-HUD systems depends on various factors, such as the vehicle's design, the driving environment, and the driver's preferences. The study suggests that the FoV of around 12 degrees horizontal and 6 degrees vertical is appropriate for automobile AR-HUD systems (Ren et al., 2016). Lastly, Xu et al. (2023) proposed a fully connected neural network (HUDNet) for an automotive AR-HUD to correct dynamic distortion. A parallel prediction framework was developed for the image's center and edge, and the distortion information was separated from the disparity information to balance the unstable central field of view error.

3.3. Projected information needed

Lee et al. (2016) introduced an automotive augmented reality 3D head-up display based on light-field rendering and eye-tracking. The study found that simpler displays led to fewer glances and improved driver performance than more complex displays, as in Lagoo et al. (2019). Multiple studies investigating safe glance durations for AR-HUD have been conducted (de Oliveira Faria, 2020; de Oliveira Faria et al., 2020; de Oliveira Faria & Gabbard, 2020; Eyraud et al., 2015; Genç et al., 2016; Kim & Gabbard, 2019; Topliss et al., 2020). Based on these papers, it is crucial not to present too much on the windshield, which could distract the driver.

The following studies provide information on different types of information that should or should not be projected on the windshield. Yoon et al. (2014) conducted a study that describes the development of an augmented in-vehicle navigation system for a HUD. The results show that the system can improve navigation accuracy and reduce driver distraction, making it suitable for in-vehicle use. Yoon et al. later proposed an assessment model for the perceived visual complexity of automotive instrument clusters (2015a) and a method for registering AR information with a head-up display (2015b). The results suggest that the proposed model can be used to optimize the design of instrument clusters for better user experience and safety. The model also improved the accuracy and consistency of AR information display in head-up displays. Bauerfeind et al. (2022) conducted a driving simulator study to examine glance behavior regarding attention allocation while participants navigated in an ambiguous left turn situation with an oncoming car in an urban area. Results showed that with the AR display, participants paid more attention to an oncoming vehicle in the ambiguous left turn situation than the HUD, indicating that distraction due to AR information was minimal. Similarly, Chauvin et al. (2023)

investigated AR-HUD navigation tasks in a complex road situation. The study showed that participants made fewer errors and drove faster on average when using AR-HUD. Moreover, [Chauvin et al. \(2023\)](#) addressed that the AR-HUD is assessed to be more useful and easier to understand than the conventional HUD.

Bolton et al. (2015) studied the effects of AR presentations of landmark-based navigation using a HUD. AR landmark-based navigation was found to be more accurate and less distracting than traditional navigation. [Bauerfeind et al. \(2021\)](#) also investigated the effects of AR on drivers' mental load while navigating. They concluded that AR navigation could reduce drivers' cognitive load by providing more natural and intuitive cues.

[Calvi et al., \(2020a\)](#) studied the effectiveness of AR warnings on driving behavior while approaching pedestrian crossings. They confirmed that augmented reality warnings significantly improved drivers' attention and reduced driving speed when approaching pedestrian crossings. They also evaluated augmented reality cues to improve the safety of left turn maneuvers in a connected environment via a driving simulator study ([Calvi et al., 2020b](#)). This study concluded that AR cues could significantly improve the safety of left turn maneuvers in a connected environment by providing better situational awareness to the driver.

[Pathivada and Perumal \(2019\)](#) analyzed drivers' behavior at signalized intersections under mixed traffic conditions. The study identified factors influencing drivers' decision-making at signalized intersections and proposed strategies to improve intersection safety. [Li et al. \(2019\)](#) also investigated the visual scanning behavior of drivers at intersections. The study was conducted in China and used naturalistic driving data. Drivers spend less time scanning the surrounding environment at signalized intersections than non-signalized ones.

[Cvahte Ojsteršek and Topolšek \(2019\)](#) examined how drivers' attention affects their ability to perceive changes in the traffic environment. The results show that visual and cognitive attention are important factors in detecting changes in the traffic environment and that cognitive load can negatively affect drivers' perception. [Li et al. \(2020\)](#) proposed a method to detect manual distraction of drivers using images of their hands and ears. The proposed method achieves a detection accuracy of 83.3 %. [Schwarz & Fastenmeier \(2018\)](#) evaluated the effectiveness of visual advisory warnings about hidden dangers using different symbols and spatial referencing methods. The results suggest that specific symbols and spatial referencing methods can reduce unnecessary warnings and increase the effectiveness of necessary warnings.

3.4. Legibility

Legibility in the human-machine interface for in-vehicle displays refers to the ease with which drivers can read and understand the information displayed on the screen while driving. As cars become more technologically advanced, the amount of information displayed on in-vehicle displays has increased, making legibility a crucial factor in ensuring driver safety. Legibility is affected by various factors such as font size, color, contrast, and layout, and designers must consider these factors when creating displays to ensure that they are easily readable while driving. Campbell et al. (2004) provide guidelines for designing icons and other information elements for in-vehicle displays. ISO 15008 provides specifications and test procedures for in-vehicle visual presentation systems ([International Organization for Standardization, 2017](#)). It covers visual design principles, human factors considerations, and FoV specifications, but most specifications may not apply to a full windshield AR-HUD.

Lastly, the following summaries are research regarding color legibility. [Gabbard et al. \(2022\)](#) proposed a color-matching method for examining the quality of color blending in AR-HUD graphics. They suggest that the AR colors of blue, green, and yellow are relatively robust, while red and brown are not. In their 2022 study, Zhong and colleagues experimented on digital characters displayed on an in-vehicle AR-HUD, using three variables: seven colors (red, green, blue, yellow, pink, yellow-red, and white), character outlines (white, grey, or none), and ambient illuminance (daytime or night). The study presented 19 stimuli to evaluate color visibility, visual fatigue, color preference, and visual intrusion. The study found that (1) yellow and green characters were most visible during the day, while red and yellow characters were most visible at night, (2) Characters with white outlines were more visible than those with grey outlines, and visibility was affected by the luminance contrast between the character and the outline (3) Non-outlined characters blended into the display background, making the luminance contrast of the character more visible. [Yamin et al. \(2023\)](#) investigated the effects of button color and background on the user experience of AR interfaces in the context of driving. The study found that the AR button color and background significantly affected the participants' accuracy and response time, with the most effective AR button color to choose to be orange and the least effective button color being blue. The orange button color had the lowest task completion time on a black and moving background. The blue button color is the least recommended compared to the other colors in this study. This contrasts with Zhou and Yin's study (2021), which suggests that using blue or green symbols can improve the usability of AR-HUDs, while red symbols may have adverse effects on usability. [Dou et al. \(2022\)](#) quantitatively analyzed the relationship between the content color and brightness of AR-HUD. A quantitative weighted color selection model was established to reflect the tendency towards selected elements to the optimal interval, providing a solution for the content color selection in AR-HUD interface design.

4. In-vehicle HMI guidelines for AR-HUD

Based on the information extracted from the literature review result in the previous section, 9 HMI guidelines were identified. These guidelines were derived from the research results and recommended in-vehicle specifications from the 104 selected articles of this review. Several studies suggested a different specification value, for example, VID ([Li et al., 2019](#); Texas Instrument, 2021). After reviewing the methods and discussion from each article, we propose the following in-vehicle HMI guidelines for AR-HUD.

4.1. Virtual image distance (VID), the field of view (FoV), and brightness

Guideline #1: Virtual image distance (VID) should be at least 6 m.

A critical feature of an AR-HUD is recognizing threats by directly marking them within the driver's field of view. To achieve this purpose, the VID requirement is at least 6 m (Texas Instruments, 2021). The previous publication (Firth, 2019) stated that 7 m is the minimum. However, it was revised to 6 m. The optimal VID is from 12 m to 15 m, and the benefit of minor misalignment of the augmented graphic from longer VIDs is diminished beyond this range (Texas Instruments, 2021).

Guideline #2: The Field of View (FoV) should be more than 10°.

Corresponding with VID, a larger field of view (FoV) will improve driver experience. A wide field of view of at least 10° is needed to present pedestrians and buildings, according to Ren et al. (2016) and Kalinina & Putilin (2020). Studies using FoV of at least 10° have been conducted, measuring driver performance and acceptance (Feierle et al., 2019, 2022; J. Lee et al., 2020, 2021) and driver risk perception and anxiety (Hwang et al., 2015, 2016). However, Ren et al. (2016) further discussed that the tracking time required depends on the field of view and varies depending on age. We propose a further study on different FoV and age groups and their effect on driving performance and the usability of AR-HUD.

Guideline #3: 10,000 nits is the minimum brightness needed when using AR-HUD in bright daytime.

AR-HUD projects graphics onto real-world objects in front of the car. Therefore, a high requirement for brightness is needed for the pictures to be seen. Based on SAE J2831 (SAE, 2012), it is recommended to have a brightness level of at least 6,000 nits to ensure visibility under different lighting conditions. In a bright ambient lighting condition, the required luminance is more than 10,000 nits (S. C. Lee et al., 2019; Y.-H. Lee et al., 2019; Zhan et al., 2020; Zhong et al., 2022). Nits are the widely used unit of luminance used by the television and display industry. It is interchangeable with the international standard unit cd/m^2 .

The weather conditions when driving will naturally change the required brightness of a display. Currently, the luminance needed for AR-HUD used indoors and outdoors is unclear. Research comparing these different environment settings and brightness levels should be done in the future.

4.2. Projected information needed

Guideline #4: In a general driving situation, AR-HUD should present information on speed, lane lines, distance from stop line, distance from other vehicles, and navigation cues. Other information should not be given.

In general, for a dynamic task like driving, the complexity of the AR-HUD display should be as simple as possible, and information should be presented when needed (Merenda et al., 2019; Yoon et al., 2015). Previous studies have highlighted speed, navigation cues, lane lines, pedestrian crossings (Calvi et al., 2020a), and distance from the stop line as important information (Pathivada & Perumal, 2019). Therefore, in Guideline #4, other information is suggested not to be presented to minimize the display complexity.

Guideline #5: When driving through an intersection, the AR-HUD should highlight the correct road lane and present spatial referencing on other vehicles and pedestrians.

Cognitive load is higher in intersections, especially in mixed traffic scenarios (Calvi et al., 2020b; Choudhary & Velaga, 2019; G. Li et al., 2019; Pathivada & Perumal, 2019). Vehicle type, a characteristic of mixed traffic conditions, significantly impacts the driver's decision at the onset of yellow.

Guideline #6: When driving at high speeds, the AR-HUD should highlight the correct road lane, distance from other vehicles, and audiovisual warning messages for collision alerts.

It is also interesting to note that highways, advertisements, and the natural environment represent the visual elements that distract the driver from taking his eyes away from the relevant traffic environment (Cvahte Ojsteršek & Topolšek, 2019). Therefore, we propose that on highways and when the vehicle is driving at high speeds, AR-HUD should increase assistance to the driver to minimize the adverse outcomes of distraction.

Guideline #7: When driving after congestion, provide audiovisual warning messages for collision alerts.

After congestion, it is found that driving behavior tends to be more aggressive and prone to accidents (Li et al., 2020). As a result, in Guideline #7, AR-HUD must increase assistive cues and warnings after congestion.

In a navigating task experiment, using augmented landmark boxes is recommended to increase the driver's attention to obstacles, compared to only using arrows (Bolton et al., 2015; Bauerfeind et al., 2021). On top of that, spatial referencing of an unspecific warning symbol consistently improved drivers' reactions (Schwarz & Fastenmeier, 2017). An animating warning message increases drivers' attention (Campbell et al., 2018). However, Schwarz and Fastenmeier (2017) found that using animation should be selected with care and only for alerting a high degree of danger.

4.3. Legibility

Guideline #8: Letter Guidelines

The text size should be 12–20 arcmin for the general readers and 30 arcmins for headings. The stroke/height ratio should be 0.08–0.2 for standard size and 0.1–0.2 for small writing. The aspect/length for alphabetic characters should be 0.65–0.8. The recommended font is San Serif, with a height/outline thickness ratio 0.08. The outline thickness for small letters should be within 0.35 mm when using an outline. Shadow direction should be 25°–155°. Stroke/shadow thickness is recommended to be 70 %. Lastly, line spacing should be three arcmins.

Relevant existing Human Factors and Human-Computer Interaction guidelines were reviewed for proposing letter and color

guidelines. Letter guidelines comply with ISO 15008 (ISO, 2017), SAE-J2831 (2012), TRL-PA3721/01 (Stevens et al., 2002), Campbell et al. (2004), and Naujoks et al. (2019). We argue that these standards apply to AR-HUD. Nonetheless, it would be worth conducting future research to validate these legibility guidelines specifically for AR-HUD and how it affects different age groups (Pampel et al., 2018).

Guideline #9: Color guidelines

Change the highlight color to red when a vehicle, bicycle, or pedestrian is less than 5 m from one's car. Use red to indicate highly urgent messages, yellow to indicate exemplary information, and green to indicate normal operations or safe conditions. Refrain from using the combinations of green–red, green–blue, yellow–red, yellow–blue, purple–red, and blue–white. Orange is recommended for an AR button or box-shaped graphic (Yamin et al., 2023). For text color, use white outlines and refrain from using grey outlines (Zhong et al., 2022). It is recommended not to use more than four colors for semantic coding (Campbell et al., 2018).

Color plays a vital role in successful interpretation. Practical uses of color can help a driver group information and code information, attract their attention more quickly, and facilitate interpretation via population stereotypes (Campbell et al., 2018).

The findings of this study's HMI design feature, recommendation, and reference are shown in Table A2. It should be noted that only studies from Table A1 were included in the guideline formulating process, but not all studies in Table A1 were included in the guidelines and Table A2.

5. Discussion

This review article aims to derive an initial guideline on AR-HUD human–machine interfaces.

5.1. Differences between traditional AR-HUD and windshield AR-HUD

AR-HUD has been mentioned in several publications about HUDs, and we found the distinction between traditional AR-HUDs and advanced windshield AR-HUD is unclear. Therefore, defining the difference between current and future AR-HUDs and early versions of AR-HUDs is meaningful. Early versions of AR-HUD also place some information on the windshield with or without putting a sticker on the windshield. We propose categorizing these devices into traditional AR-HUD because the projected information, such as tracking pedestrians and nearby vehicles, could not interact with the real world. Based on the results found in section 4, we could distinguish that, unlike traditional AR-HUDs, where the graphics do not overlay onto real-world objects, the definition of a true AR-HUD should. Fig. 2 depicts the requirement differences between traditional AR-HUDs and windshield AR-HUDs. Fig. 3 shows a traditional AR-HUD (right) and windshield AR-HUD (left).

Another notable difference between a traditional AR-HUD and an advanced AR-HUD is where the information is placed. Traditional AR-HUDs are usually located above, behind, or left/right of the steering wheel. The further away it is from the road line of sight, the

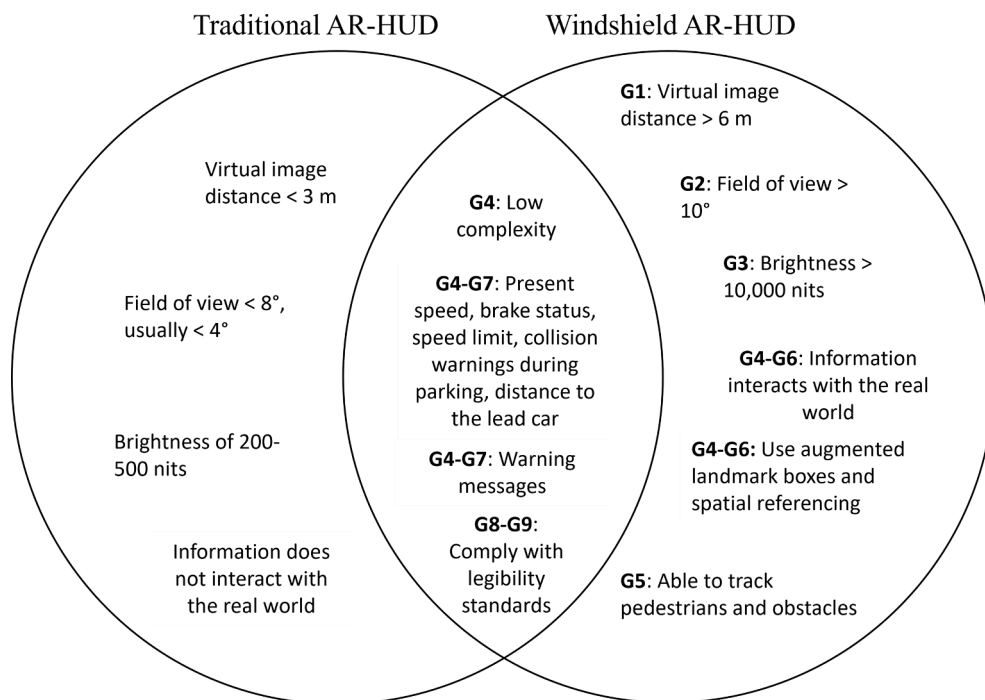


Fig. 2. Comparison of traditional AR-HUD vs. windshield AR-HUD requirements. The letter G followed by a number indicates the relevant guideline number proposed in this study (see Section 4).



Fig. 3. Example illustrations of a windshield AR-HUD (left) and conventional AR-HUD (right), used in Bellet et al. (2018).

longer the duration of the driver's glance at the HUD. An in-vehicle HUD is usually located 40–60 cm from the driver's eyes, whereas earlier AR-HUD projects its image to 2–3 m in front of the driver (Maag et al., 2023). Because a conventional AR-HUD image is located near the driver, it is adequate for the field of view to be below 8° , even less than 4° for most devices. A modern AR-HUD, however, should project the augmented graphic on the windshield and onto real-world objects in the driver's line of sight while driving. This means augmented landmark boxes, lines, and text are projected on real-world objects like roads, buildings, vehicles, and pedestrians. Therefore, the driver will not need to move their eyes from the road to see the necessary information.

To accommodate all these features, the virtual image distance should be at least 6 m, with 12–15 m as the optimal distance, and the field of view should be more than 10° (Texas Instruments, 2021). When the VID is greater than 6 m, the eyes are less sensitive to discrepancies in physiological distance cues and will perceive that the AR information is more strongly fused with the real world.

A longer VID in an AR-HUD system can result in a better user experience and lower system cost. A longer VID will result in less eye fatigue, a negligible parallax effect when the driver moves their head or eyes, a better fusion between AR information and real-world objects, and eliminating wedge film and eye-tracking requirements. In their technical white paper, Texas Instruments (2021) states that the optimal VID is between 12 m and 15 m. In this range, virtual images are possibly fused with the real world. However, the benefit diminishes if the VID increases beyond this recommended range. It must be considered that a longer VID negatively impacts HUD systems with traditional optical designs by creating a solar load challenge. Therefore, technology that can tolerate much higher solar loads and enable long VIDs without thermal issues is needed.

The AR-HUD should also accurately mark surrounding vehicles and pedestrians in night vision/rain-fog conditions, improving driving safety in poor visibility. The brightness and contrast of the virtual image are critical to achieving this. Traditional HUDs are inside the vehicle with a still background, so the brightness level required is around 200–500 nits. However, the virtual image from

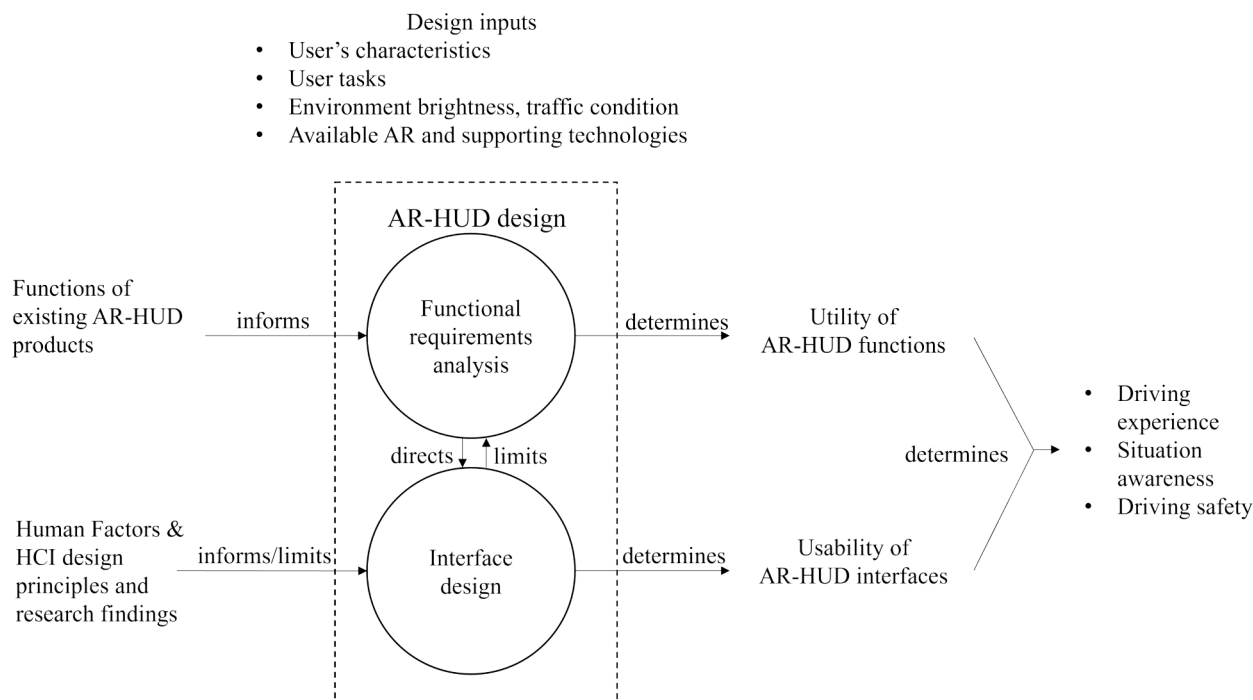


Fig. 4. AR-HUD design conceptual framework.

advanced AR-HUDs is projected into the outdoor environment. Therefore, the brightness level required is the same as that required when using AR outdoors during bright daytime, which is 10,000 nits (Lee et al., 2019a, 2019b; Zhan et al., 2020).

What and when certain warning information should be presented is essential. Guidelines 4–7 discuss this. Current advanced non-AR-HUDs use audiovisual warning messages. The difference between AR-HUDs is that the image could now present augmented landmark boxes and spatial referencing, further improving the driver's attention to target objects.

5.2. AR-HUD design conceptual framework

Park and Park (2019) reviewed the literature on head-up displays in automotive applications to identify the functional requirements of such displays. The study identifies the key functional requirements of head-up displays, including visibility, readability, and usability, and provides recommendations for the design of such displays. Based on the literature review and proposed AR-HUD guidelines, a conceptual framework has been formulated to further guide the development of AR-HUD design. Fig. 4, based on Park and Park's (2019) HUD framework, depicts the conceptual framework that shows the relationship between the design inputs, functional requirement analysis, and interface design for AR-HUD and how it affects usefulness in driving. AR-HUD design consists of available requirement analysis and interface design. The functional requirement analysis results in direct interface design, whereas interface design limits the functional requirement. Similarly, design inputs such as user characteristics, tasks, environment conditions, and technology availability (and limitation) are factors considered in AR-HUD design. Strle et al. (2023) investigated physiological indicators for modeling HUD-induced cognitive load to understand the relationship between the user's characteristics and HUD type. Tabone et al. (2023) highlighted the importance of involving target users in design and evaluation.

Functional requirement analysis is created from information derived from functions of existing AR-HUD and mapping the user needs. It is then used to determine the AR-HUD functions which will affect utility. In contrast, interface design is informed and limited by existing Human Factors & Human-Computer Interaction design principles and guidelines. This will create the AR-HUD interface, which affects its usability. In turn, the usability of the AR-HUD interface determines the driving experience (Liang et al., 2023), driver situation awareness (see Greatbatch et al. (2022) and Mukhopadhyay et al. (2023), and driver safety (Maag et al., 2023). This is in line with numerous studies. Yontem et al. (2021) proposed HMI provides a more immersive and intuitive driving experience, improving driver safety and comfort. Calvi et al., (2020a) and Gabbard et al. (2019) also discuss the link between AR-HUD interface usability and positive driving experience (Halit et al., 2021). Through an eye-tracking study, Cheng et al. (2023) verified that AR-HUD technology can significantly improve the driver's attention to risky Areas of Interest. It has been shown that the AR-HUD interface can reduce the difficulty of processing information in risky driving scenarios.

5.3. Key results and implications of this study

The novel contributions of this article are the following: (1) Providing a systematic overview of existing research associated with the human-vehicle interface for AR-HUD (Table A.1) (2) Formulating a set of guidelines for the human-vehicle interface to develop AR-HUD (Table A.2) (3) Differentiating traditional versions of AR-HUD with windshield AR-HUD (Fig. 2) (4) Provided an AR-HUD design conceptual framework (Fig. 4) (5) Outlining the key areas where future research on AR-HUD is needed.

Fig. 5 depicts the summary of the proposed guideline formulated in this study. Future windshield AR-HUD should be developed

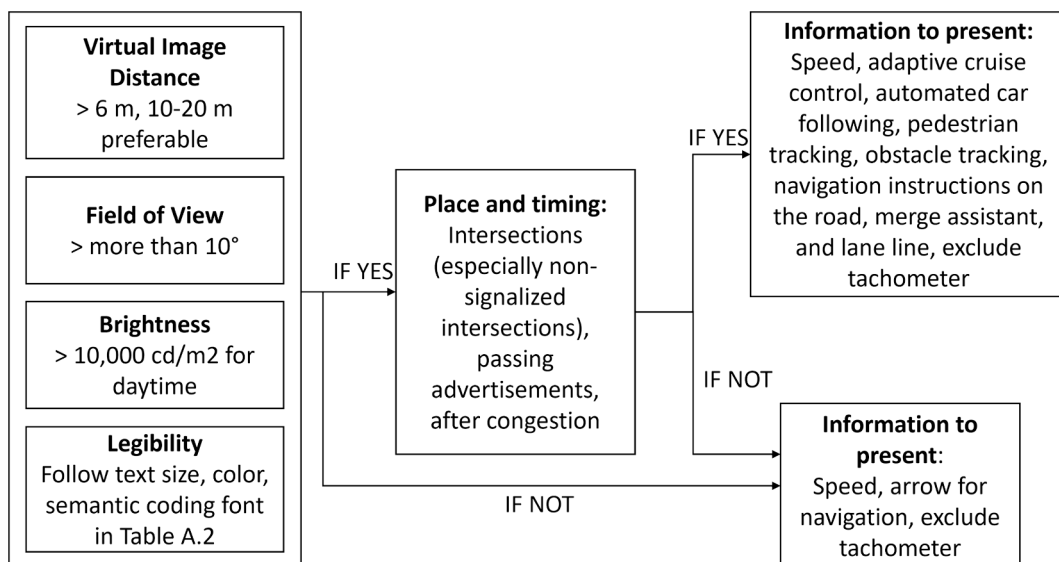


Fig. 5. Key findings of the literature review.

following the VID, FoV, Brightness, and legibility requirements as stated in the left boxes. Only with these specifications and at critical places and times, such as intersections, passing advertisements, and after congestion, should the AR-HUD present a higher amount of information. If not, it is recommended only to show very simple information such as speed and arrow for navigation. An in-vehicle AR-HUD aims to improve the driving experience and safety (Gabbard et al., 2019). The design of AR-HUD should ensure that the device aids the driver instead of increasing distraction by displaying excessive information (Naujoks et al., 2015; Eyraud et al., 2015) and is accepted and trusted by the driver (Li et al., 2022).

There is no one-size-fits-all recommendation for the human-machine interface (HMI) specification for an in-vehicle augmented reality head-up display (AR-HUD), as the optimal HMI design depends on various factors such as the specific use case, user preferences, and technological constraints. However, some general principles for designing an effective AR-HUD include ensuring that the display is well-integrated with the driver's field of view, providing relevant and timely information, minimizing distraction and cognitive workload, and offering customizable features to cater to individual user needs. Additionally, the HMI should be designed with safety in mind, ensuring that the driver's attention is not diverted from the road and that the display does not cause visual or cognitive overload. Ultimately, an AR-HUD HMI aims to enhance driving performance and situational awareness while minimizing the risk of accidents or errors.

The results of this study summarize essential human-machine interface features to achieve the intended benefits of AR-HUD. The study is expected to be beneficial for furthering the design of an AR-HUD-assisted system. However, there will be many challenges that need to be addressed for an AR-HUD designer and manufacturer.

First is graphics alignment or sensor fusion. It is displaying the graphics on the correct real-world object based on the vehicles' sensor data (Chen et al., 2023). The rapid change in the environment's movement, color, and brightness would mean the AR-HUD HMI software should be able to decide and adapt to all changes. The position of the driver's eye box also contributes to the graphics alignment challenge, called the parallax effect. Parallax is the change of alignment of the virtual image and the real-world object when the human eye changes position. When a driver moves their head slightly to one side, the virtual image will also move and cause misalignment. Chen et al. investigated the impact of AR graphic position (peripheral AOI v.s. central AOI) and the relative position of the AR graphic on unpredictable stimuli (on-HUD hazard v.s. off-HUD hazard) on the occurrence of inattention blindness. In the peripheral visual field, unpredictable stimuli accidentally superimposed on the AR graphic (i.e., on-HUD hazard) lead to a higher probability of ignoring the accidental events and seem to require a longer response time for drivers. The study illustrated that inattention blindness to nonaugmented stimuli is also influenced by the AR graphic position when AR technology fails to augment them promptly. An important recommendation emerging from this work is to consider the design of AR graphics according to the AR graphic positions and stimulus types to reduce the occurrence of inattention blindness. Furthermore, Texas Instruments (2021) calculated the magnitude of misalignment compared to different VID and eye positions. When the VID is at 13 m, the misalignment is negligible.

Accommodating differing driver heights requires adjusting the eye-box size. Regardless of the type of technology used, a larger eye box comes with the expense of increased lumens. The lumens required to support a HUD design are directly proportional to the FOV and eye-box area. The greater the number of lumens, the greater the power consumption and the more challenging the thermal design becomes (Firth, 2019). Further research on developing efficient imaging technology for all users and developing the complete AR-HUD software should be done.

An AR-HUD needs to produce constant performance in brightness, color, latency, and contrast in any temperature, driving condition, and power consumption. Furthermore, longer VIDs and associated higher magnification create significant solar load challenges. Traditional HUDs have optical designs with a magnification of around five times, whereas the AR-HUDs have a magnification of 25x to 30x. It is critical to choose an efficient light source and imaging technology to support the required lumen budget at minimal power consumption while considering the thermal implications. Further research should be conducted on the relationship between FoV, eye-box size, brightness level, and lumen.

Early versions of AR-HUD have a challenge with polarization. In some imaging technologies, a driver wearing polarized sunglasses could not see the HUD image. The information from the AR-HUD is essential. Therefore, advanced AR-HUD must tackle this problem and make sure all images are visible when wearing any glasses.

5.4. Limitations and future work

This study comes with the following limitations that leave room for further research. This study did not develop an AR-HUD prototype, nor did we use a commercial AR-HUD to investigate. The focus of this paper was to present a set of the most relevant guidelines that can be used for designing AR-HUD. Furthermore, this study did not consider the costs and limitations of the current technology available. This is because this review was intended to provide information on the near-optimal HMI specification and features. Advancements in technology and the widespread use of this product will eventually lower the production cost and price. Nevertheless, future studies should be conducted on the willingness to purchase and use AR-HUD.

We propose a further study on different FoV and age groups and their effect on driving performance and the usability of AR-HUD. Bellet et al. (2018) raised the issue of difficulties experienced by older drivers with advanced driving aid systems such as AR-HUD. Brightness (luminance) is compromised when the environmental luminance changes; however, the difference in brightness values required has not been studied yet. Therefore, future studies should be performed on this. Next, the luminance needed for AR-HUD used indoors and outdoors is unclear. Research comparing these different environment settings and brightness levels should be done in the future.

The following are more recommendations for future AR-HUD studies. There is a need for a study on the costs and use of AR-HUD. Second, a study that includes a control group will compare the driving performance and user experience with and without AR-HUD

technology. Third, future experiments should incorporate a diverse range of driving scenarios, such as highway driving, urban driving, and adverse weather conditions, to assess the effectiveness of AR-HUD systems in various situations. Third, the long-term effects of using AR-HUD technology on driver behavior and safety over an extended period should be evaluated. Fourth, a comparative analysis of different AR-HUD designs and features will be conducted to determine the most effective configuration for improving driving performance and user satisfaction. Fifth, future studies should consider the impact of drivers' age and experience level on the usability and acceptance of AR-HUD systems. Lastly, a future work should be done on validating legibility guidelines specifically for AR-HUD and how it affects different age groups (Pampel et al., 2018).

6. Conclusions

The current study derived initial guidelines for developing AR-HUD from existing literature and applicable guidelines and standards. In total, 121 key articles relevant to this study were selected in this review, consisting of 74 journal papers, 40 conference proceedings, two technical reports, and five Human Factors and Human-Computer Interaction Guidelines. The major findings of this study are the following. Firstly, we analyzed the development of AR-HUD and categorized the articles based on HMI features relevant to in-vehicle AR-HUD: virtual image distance, the field of view, and brightness. Furthermore, research findings on projected information and legibility were investigated. Future windshield AR-HUD is recommended to be developed following the VID, FoV, brightness, and legibility requirements. Only with these specifications and at critical places and times, such as intersections, passing advertisements, and after congestion, should the AR-HUD present a higher amount of information. If not, it is recommended only to show elementary details such as speed and arrow for navigation. Secondly, an AR-HUD design conceptual framework was made based on the findings. An in-vehicle AR-HUD aims to improve the driving experience, driver's situational awareness, and safety. The AR-HUD design should ensure that the device helps the driver, not distracts it. Thirdly, this study also compared the traditional AR-HUD and windshield AR-HUD, defining the requirements for future studies using the term AR-HUD.

It has been found that current research on AR-HUD is limited. Based on the findings, future research is needed on the following: Experiment on different VIDs to drive performance, investigate the relationship between other FoVs and other age groups to AR-HUD usability and operating performance, investigate the costs of AR-HUD technology, evaluate the long-term effects of using AR-HUD technology on driver behavior and safety over an extended period, comparing different environment settings and different brightness levels, and to validate legibility guidelines specifically for AR-HUD.

CRedit authorship contribution statement

Putra A.R. Yamin: Writing – original draft. **Jaehyun Park:** Supervision. **Hyun K. Kim:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. NRF-2021R1A2C4002641).

Appendix

Table A1
Summary of data extracted from selected articles on in-vehicle hmi for ar-hud.

No.	Reference	Independent Variables	Measures	VID	FoV	Brightness	Projected Information	Legibility
1	Broy et al. (2014)	–	3D head-up display design parameters, user perception	v	v	v		
2	Lisle et al. (2019)	Depth cues in volumetric AR displays	Human depth perception accuracy and precision	v	v		v	
3	Lv et al. (2021)	Volume holographic optical element technology	Multi-plane AR-HUD functionality and performance	v	v	v		

(continued on next page)

Table A1 (continued)

No.	Reference	Independent Variables	Measures	VID	FoV	Brightness	Projected Information	Legibility
4	Pullukat et al. (2020)	image distance	Occurrence of cognitive tunneling	v				
5	Richter et al. (2018)	Holographic waveguide augmented reality head-up display	Visual quality, the field of view, and user satisfaction	v	v	v		
6	Firth (2019)	–	–	v	v	v		
7	Texas Instruments (2021)	Vertical Image Distance (VID)	Driver performance, comfort, and workload	v	v			
8	Wang et al. (2018)	Digital holographic system for automotive augmented reality head-up display	Image quality, the field of view, and user satisfaction	v	v	v		
9	Abdi et al. (2015)	–	User satisfaction, the accuracy of driving, and traffic information		v		v	
10	Bauerfeind et al. (2021)	Type of navigation system	Mental workload		v		v	
11	Bremers et al. (2021)	Head-up display types, type of visual representation	Perception of depth and perspective in augmented reality displays		v	v		
12	Calvi et al., (2020b)	Presence of augmented reality cues	Driver response and accuracy in left-turn maneuvers		v		v	
13	Choudhary & Velaga (2019)	Distraction (phone and music player)	Gap acceptance behavior at unsignalized intersections		v		v	
14	Cvahte Ojsteršek and Topolšek (2019)	Visual and cognitive attention, traffic environment changes	Perception of traffic environment changes		v		v	
15	de Oliveira Faria et al. (2020)	User interface type	Drivers' spatial knowledge acquisition and glance behavior		v		v	
16	de Oliveira Faria (2020)	Presence of an augmented reality head-up display	Driver performance and distraction		v	v		
17	Eyraud et al. (2015)	Allocation of visual attention	Driving performance with simulated augmented reality		v		v	
18	Feierle et al. (2019)	Presence of augmented reality	Driving performance and acceptance		v			v
19	Feierle et al. (2022)	Presence of an augmented reality head-up display	Driver performance and acceptance during malfunctions		v			
20	Gabbard et al. (2019)	Immersive driving simulator	Evaluation of augmented reality head-up display effects on driving performance and distraction		v			v
21	Gabbard & Mehra (2019)	AR display context and focal distance	Human performance		v		v	
22	Gabbard et al. (2022)	Color-matching method	Perception of color blending in augmented reality head-up display graphics		v			
23	Gao et al. (2019)	Calibration method	Calibration accuracy for automotive augmented reality head-up displays		v			
24	Genç et al. (2016)	Head-mounted projection display	Visual attentional processing of head-referenced displays		v			
25	Halit et al. (2021)	Use of augmented reality head-up-display (AR-HUD)	Driving performance, situation awareness, and cognitive workload		v			
26	Hwang et al. (2016)	Use of augmented reality head-up-display (AR-HUD)	Risk perception, psychological changes of drivers		v			v
27	ISO 15,008 (2017)	–	Standards for in-vehicle visual presentation		v			
28	Kalinina et al. (2020)	Type of waveguide with pupil replication used in the AR head-up display	Color accuracy, brightness, contrast, and field of view		v	v		
29	Kim and Gabbard (2019)	Use of augmented reality head-up-display (AR-HUD)	Distraction potential of AR-HUDs		v	v	v	
30	Kim et al. (2013)	Use of head-up-display (HUD) and augmented reality (AR)	Crash warning system performance		v		v	
31	Kim et al. (2016)	Use of augmented reality head-up-display (AR-HUD)	Pedestrian warning system performance		v			
32	Kim et al. (2016)	Use of augmented reality head-up-display (AR-HUD)	Cross-traffic dynamics visibility		v		v	
33	Lagoo et al. (2019)	Use of head-up-display (HUD) and gesture recognition system (GRS)	Distraction mitigation		v		v	

(continued on next page)

Table A1 (continued)

No.	Reference	Independent Variables	Measures	VID	FoV	Brightness	Projected Information	Legibility
34	Lee et al. (2016)	The visual complexity of in-vehicle information display	Glance behavior and preferences		v		v	
35	Lee et al. (2019)	–	Discussion of prospects and challenges in augmented reality displays		v		v	
36	Lee et al. (2020)	Use of augmented reality 3D head-up-display (AR-HUD) based on light-field rendering with eye-tracking	Driving performance, eye movements		v	v		
37	Lee et al. (2021)	Use of augmented reality head-up-display (AR-HUD)	Information visualization enhancement		v	v		
38	Li et al. (2019)	Scanning behavior of drivers at signalized and unsignalized intersections	Scanning behavior of drivers		v		v	
39	Li et al. (2022)	User demographics, AR-HUD technology, connected vehicle features	User acceptance and intention to use AR-HUD		v		v	
40	Ma et al. (2021)	AR-HUD display type, driving task	Driving performance and eye movements		v			
41	Maag et al. (2023)	Type of head-up display (AR-HUD vs. conventional HUD)	Workload (detection response task)		v			
42	Merenda et al. (2018)	AR interface design, task type	User task performance and workload		v		v	
43	Merenda et al. (2019)	AR-HUD visual fidelity, driving task	User task performance and subjective experience		v		v	
44	Murugan et al. (2022)	Autonomous vehicle technology, AR-HUD design, machine learning algorithms	User experience and performance in autonomous vehicle operation		v	v		
45	Nachiappan et al. (2021)	AR-HUD display type, driving task	Driving performance and user experience		v		v	
46	NHTSA (2018)	Human factors design considerations, level 2 and 3 automated driving concepts	Design guidance for human factors considerations in level 2 and 3 automated driving concepts		v			
47	Okumura et al. (2018)	Optical combiner design	Performance or effectiveness of the augmented reality head-up display		v			
48	Pampel et al. (2018)	User age, navigation system type	User performance and experience with navigation systems in AR-HUD		v		v	
49	Park et al. (2015)	Augmented reality technology, driving conditions	User performance and experience driving at night with AR technology		v			
50	Park et al. (2015)	AR technology, driving situation awareness	User performance and situation awareness with AR technology		v			
51	Park et al. (2016)	AR-HUD design, vehicle information	User experience and performance with AR-HUD and vehicle information		v			
52	Park and Im (2020)	HMI design guidelines, semi-automated driving	User experience and performance with AR-HUD in semi-automated driving		v			
53	Park and Park (2019)	HUD technology, functional requirements	Summary of functional requirements for AR-HUD in automotive applications		v			
54	Pathivada and Perumal (2019)	Signalized intersection design, mixed traffic conditions	Driver behavior and decision-making at signalized intersections		v		v	
55	Pauzie (2015)	AR-HUD technology, automotive application	User experience and acceptance of AR-HUD in automotive applications		v	v		
56	Pffannmüller et al. (2015a)	Navigation system display concept	Performance of drivers in terms of speed, accuracy, and mental workload		v		v	
57	Pffannmüller et al. (2015b)	Depth perception of augmented reality information	Accuracy of depth perception		v	v		
58	Poitschke et al. (2008)	Contact-analog information representation	User satisfaction, ease of use, and performance		v	v	v	
59	Qin et al. (2019)	Dual-focal-plane augmented reality head-up display	Optical performance, visual comfort, and user satisfaction		v	v	v	
60	Ren et al. (2016)	Wide-field-of-view augmented reality	User experience, task completion time, and accuracy		v			

(continued on next page)

Table A1 (continued)

No.	Reference	Independent Variables	Measures	VID	FoV	Brightness	Projected Information	Legibility
61	SAE-J283 (2012)	Alphanumeric message design and engineering recommendations	Driver comprehension and response time		v		v	v
62	Schömig et al. (2018)	Augmented reality display	User perception, trust, and performance		v			
63	Smith et al. (2016)	Head-up and head-down displays	Driver performance, safety, and subjective workload		v		v	
64	Smith et al. (2017)	Augmented reality head-up displays	Eye scan patterns, driving performance, and subjective workload		v			
65	Smith et al. (2020)	Augmented reality head-up display	Driver response time, accuracy, and workload in unexpected events		v		v	
66	Tonniss et al. (2007)	Longitudinal and lateral driving assistance	Driving performance, subjective workload, and user satisfaction		v			
67	TRL-PA3721/01 (Stevens et al., 2002)	In-vehicle information system design guidelines	Driver safety and usability		v			
68	Campbell et al. (2004)	–	Design guidelines for in-vehicle display icons and other information elements			v		
69	Hwang et al. (2015)	Use of augmented reality head-up-display (AR-HUD)	Perception of precautionary situations			v		v
70	Qin et al. (2017)	Ghost image density (expressed in terms of the number of ghost images per unit area)	Maximal acceptable ghost image density			v		v
71	Zhan et al. (2020)	Augmented reality and virtual reality displays	Perspectives and challenges related to the use of these displays			v	v	
72	Zhong et al. (2022)	Seven colors (R, G, B, Y, P, YR, W) of character, the outlines (white, grey, none) of character, and ambient illuminance (daytime, night)	Color visibility, visual fatigue, color preference, and visual intrusion			v		v
73	Zhou and Yin (2021)	Symbol color of automotive augmented reality head-up display	Effectiveness of symbol color in automotive augmented reality displays			v		v
74	Betancur et al. (2018)	–	Trends and developments in automotive head-up display systems				v	
75	Bolton et al. (2015)	The presentation mode of navigation information	Navigation performance, driver preference				v	
76	Calvi et al., (2020a)	Presence of an augmented reality warning system	Driver response to pedestrian crossings				v	
77	Charissis (2014)	–	Driver response time, accuracy, and workload				v	
78	de Oliveira Faria & Gabbard (2020)	Duration of glance	Safety and usability of augmented-reality head-up display				v	
79	Greatbatch et al. (2022)	Three different types of AR graphics: no AR graphics, basic AR graphics, and advanced AR graphics	Driver's situation awareness and takeover performance (self-reported situational awareness, reaction times, and performance on a driving simulator task)				v	
80	Kim and Hwang (2017)	Use of augmented reality head-up-display (AR-HUD)	Usefulness of AR-HUDs				v	
81	Langlois and Soualmi (2016)	Type of head-up display (augmented reality versus classical)	Time to react to a takeover request, the accuracy of the takeover maneuver, and the time to complete the maneuver				v	
82	Li et al. (2020)	Hand and ear movements and positions	Detection of driver distraction				v	
83	Naujoks et al. (2015)	Cooperative perception system, warning types	User response and performance based on warning types				v	
84	Naujoks et al. (2019)	Automated vehicle technology, HMI design guidelines	Guidelines and verification methods for automated vehicle HMIs				v	
85	Schwarz and Fastenmeier (2017)	Advisory warning symbols and spatial referencing	Driver response time and accuracy				v	v

(continued on next page)

Table A1 (continued)

No.	Reference	Independent Variables	Measures	VID	FoV	Brightness	Projected Information	Legibility
86	Soomro and Urey (2020)	Presence of accommodation-convergence conflict	Landolt C visual acuity test targets presented on the 3D head-up display				v	
87	Tonniss et al. (2005)	Augmented reality visualization	Driver response time and accuracy				v	
88	Topliss et al. (2019)	Lead vehicles as driving navigational aids	Driving performance and user satisfaction				v	
89	Topliss et al. (2020)	Duration of head-up display (HUD) presentation	Driver's ability to detect a peripheral event while looking at the HUD				v	
90	Tran et al. (2013)	Augmented reality left-turn driving aid	Driver response time and accuracy in left-turn situations				v	
91	Utsuboya et al. (2013)	Augmented reality head-up display	Usability and user satisfaction in-car navigation system				v	
92	Valimont et al. (2007)	Augmented reality information acquisition	Task completion time, accuracy, and user satisfaction in aviation maintenance				v	
93	von Sawitzky et al. (2019)	Augmented reality head-up display for fully automated driving	User perception of trust and reliability				v	
94	Wang et al. (2019)	Augmented reality head-up display system for rear passengers	Reduction in driver distraction				v	
95	Wang et al. (2022)	Augmented reality head-up display	Detection of inattentional blindness during driving				v	v
96	Wei et al. (2019)	Design of a head-up display system	Freeform reflective system for automotive applications				v	
97	Weinberg et al. (2011)	Head-up display for selection from choice lists	Usability				v	
98	Wientapper et al. (2013)	Camera-based calibration system for augmented reality head-up displays	Accuracy of calibration				v	
99	Yontem et al. (2021)	Immersive human-machine interface, multiple zones, full windshield, head-up display	Prospective analysis of a new human-machine interface for vehicles				v	
100	Yoon et al. (2014)	The augmented in-vehicle navigation system	Development of a navigation system for head-up display				v	
101	Yoon et al. (2015)	The visual complexity of automotive instrument cluster	Assessment model for perceived visual complexity				v	
102	Yoon and Kim (2015)	Augmented reality information registration for head-up display	Accuracy of augmented reality information registration				v	
103	Zhang et al. (2021)	Full windshield head-up display or traditional instrument cluster	Visual attention allocation of the driver				v	
104	Yamin et al. (2023)	Background color, AR button color	Task completion time, number of errors, button color visibility					v
105	Abu-Khadrah et al. (2023)	Development of adaptable recommendation system	—				v	
106	Bauerfeind et al. (2022)	Traffic complexity	Glances, NDRT, subjective rating				v	
107	Bellet et al. (2018)	Navigation, speed control, intersection crossing, etc.	Questionnaire –focus group				v	
108	Chauvin et al. (2023)	Three navigation tasks	Driving error, driving speed, subjective ratings				v	
109	Chen et al. (2023)	Graphic position, AR graphic position on unpredictable stimuli	Rate of inattentional blindness		v			v
110	Cheng et al. (2023)	Risky driving scenarios, daytime/nighttime	Attention to AOI, reaction time			v	v	
111	Dou et al. (2022)	Color, brightness	Visual sensitivity, fatigue			v		v
112	Gabbard et al. (2022)	Color robustness, backgrounds	Shift, dispersion					v
113	Greatbatch et al. (2023)	Visual graphic types	Situation awareness, takeover performance				v	
114	Liang et al. (2023)	Novel visual mechanism	Situation awareness, driving experience				v	
115	Liu et al. (2023)	AR scenarios, experiential prototypes	User experience questionnaire				v	
116	Maag et al. (2023)	Three display technologies	detection response task. Gaze behavior, subjective				v	

(continued on next page)

Table A1 (continued)

No.	Reference	Independent Variables	Measures	VID	FoV	Brightness	Projected Information	Legibility
117	Mukhopadhyay et al. (2023)	AR and MR driver assistance prototypes	Response time to TOR				v	
118	Sburlan et al. (2020)	Head position, development of driver's head monitoring module	Interaction time	v	v			
119	Strle et al. (2023)	HUD type: baseline HUD and AR-HUD	heart rate variability, electrodermal activity, skin temperature, pupil dilation				v	
120	Tabone et al. (2023)	AR interfaces	Online questionnaire				v	
121	Xu et al. (2023)	Proposed calibration method	Dynamic distortion		v			

Table A2

In-vehicle HMI guidelines for AR-HUD.

Interface design feature		Recommendation	Reference
Information	Virtual Image Distance	more than 6 m, 10 to 20 m preferable	Texas Instruments (2021)
	Field of View	more than 10°	Texas Instruments (2021), Ren et al. (2017)
	Brightness	more than 10,000 cd/m ² for daytime and outdoors	Lee et al. (2019a, 2019b), Zhan et al. (2020)
	Complexity	Low complexity: speed, arrow for navigation, exclude tachometer	Yoon, et al. (2015), Lee et al. (2016)
	Information to present	Adaptive cruise control, automated car following, pedestrian tracking, obstacle tracking, navigation instructions on the road, merge assistant, and lane line	Merenda et al. (2018), Bolton et al. (2015), Pathivada and Perumal (2019), Choudary and Velaga (2019)
	Where & when to increase assisting information	Intersections (especially non-signalized intersections), advertisements, after congestion	Choudary and Velaga (2019), Pativada and Perumal (2019), Cvahte Ojstersek & Topolsek (2019), Li, et al. (2019), Li et al. (2020)
	Availability	Graphic always on	Kim et al. (2019)
	Visual cues	Use augmented landmark boxes	Bolton et al. (2015)
		Use spatial referencing	Shwarz and Fastenmeier (2018)
	Short sentence	Emergency (within 3 words), while driving (within 4 words)	ISO-15008 (2017), SAE-J2831 (2012), Campbell et al. (2018), Yoon, et al. (2015), Naujoks et al. (2019)
	Long sentence	Normal (within 2 lines)	
	Modality	Urgent/critical (audiovisual concurrent)	
	Emphasis	Target size, state change (e.g., flashing)	
	Forward visibility time	2 s on average, 12 s in total	
	Forward non-monitor ratio	Within 15 % of the non-monitoring ratio for more than 2 s	
Letter	Array of information	The main information in the beginning/front	
	Size	Title (30'), General (range (12'–20')), General (appropriate) (16') (*Arcmin(') units can be used after changing to radian)	Campbell et al. (2004), Yoon et al. (2015)
	Stroke/Height Ratio	Normal (0.08—0.2) and Small (0.1—0.2)	
	Aspect/length/length ratio	alphabetic characters (0.65 to 0.8)	Naujoks et al. (2019), Yoon, et al. (2015)
	Font	Generic (all recognizable characters), recommended (San Serif)	ISO-15008 (2017), SAE-J2831 (2012), TRL-PA3721/01 (Stevens, 2002), Yoon et al. (2015)
	Height/Outline	San Serif (within 0.08), Serif (within 0.04)	Yoon et al. (2015)
	Thickness Ratio		
	Outline Thickness	Small letters (within 0.35 mm)	Campbell et al. (2004)
	Shape	Important content (horizontal ↑, thickness of stroke ↑, outline)	
	Shadow Direction	25~155 (* Based on 0° vertical top)	
	Stroke/Shadow	70 % in no time	
	Thickness		
	Line spacing	3'	
	Wavelength	450 nm Prohibit before and after (highly saturated blue)	Campbell et al. (2004), Naujoks et al. (2019), Zhong et al. (2022), Yamin et al. (2023)
	Background-Target	Background (dark system), letters/symbols (bright system, orange)	
Color	Background-button-letter Color	General (Black-grey-White), if preferred by individuals (Black-Blue-White)	
	Text color	Yellow and green for daytime, red and yellow for nighttime	
	Text outline color	Use white outline, refrain from grey outline	
	Refrain from using a combination	Green-red, green-blue, yellow-red, yellow-blue, purple-red, blue-white	
	Background-Destination	Normal (3:1), Night Mode (5:1)	Naujoks et al. (2019)
	Semantic coding	Red (hazard), Yellow (warning/care), Green (normal)	Campbell et al. (2004)
		Fatal (black, red), green (safe, go, increase, start), red (stop, fatal)	Campbell et al. (2004)
	Type of semantic coding	Recommended to use no more than 4	Naujoks et al. (2019)

References

- Abdi, L., Abdallah, F. B., & Meddeb, A. (2015). In-vehicle augmented reality traffic information system: A new type of communication between driver and vehicle. *Procedia Comput. Sci.*, 73, 242–249. <https://doi.org/10.1016/j.procs.2015.12.024>
- Abdi, L., & Meddeb, A. (2018). Driver information system: A combination of augmented reality, deep learning and vehicular Ad-hoc networks. *Multimed. Tools Appl.*, 77(12), 14673–14703. <https://doi.org/10.1007/s11042-017-5054-6>
- Abu-Khadrah, A., Jarrah, M., Alrababah, H., Alqattan, Z. N. M., & Akbar, H. (2022). Pervasive computing of adaptable recommendation system for head-up display in smart transportation. *Comput. Electr. Eng.*, 102, Article 108204. <https://doi.org/10.1016/j.compeleceng.2022.108204>
- Bauerfeind, K., Driike, J., Schneider, J., Haar, A., Bendewald, L., & Baumann, M. (2021). Navigating with Augmented Reality – How does it affect drivers' mental load? *Appl. Ergon.*, 94, Article 103398. <https://doi.org/10.1016/j.apergo.2021.103398>
- Bauerfeind, K., Driike, J., Bendewald, L., & Baumann, M. (2022). How does navigating with Augmented Reality information affect drivers' glance behaviour in terms of attention allocation? *Frontiers in Virtual Reality*, 3, Article 930117. <https://doi.org/10.3389/frvir.2022.930117>
- Bellet, T., Paris, J.-C., & Marin-Lamellet, C. (2018). Difficulties experienced by older drivers during their regular driving and their expectations towards Advanced Driving Aid Systems and vehicle automation. *Transport. Res. F: Traffic Psychol. Behav.*, 52, 138–163. <https://doi.org/10.1016/j.trf.2017.11.014>
- Bremers, A. W. D., Yönten, A.O., Li, K., Chu, D., Meijering, V., & Janssen, C. P. (2021). Perception of perspective in augmented reality head-up displays. *Int. J. Hum. Comput. Stud.*, 155, Article 102693. <https://doi.org/10.1016/j.ijhcs.2021.102693>
- Broy, N., Höck, S., Frederiksen, A., Gilowski, M., Eichhorn, J., Naser, F., Jung, H., Niemann, J., Schell, M., Schmid, A., & Alt, F. (2014). Exploring design parameters for a 3D head-up display. *Proceedings of The International Symposium on Pervasive Displays*, 38–43. <https://doi.org/10.1145/2611009.2611011>
- Calvi, A., D'Amico, F., Ferrante, C., & Bianchini Ciampoli, L. (2020a). Effectiveness of augmented reality warnings on driving behaviour whilst approaching pedestrian crossings: A driving simulator study. *Accid. Anal. Prev.*, 147, Article 105760. <https://doi.org/10.1016/j.aap.2020.105760>
- Calvi, A., D'Amico, F., Ferrante, C., & Bianchini Ciampoli, L. (2020b). Evaluation of augmented reality cues to improve the safety of left-turn maneuvers in a connected environment: A driving simulator study. *Accid. Anal. Prev.*, 148, Article 105793. <https://doi.org/10.1016/j.aap.2020.105793>
- Campbell, J. L., Brown, J. L., Graving, J. S., Richard, C. M., Lichty, M. G., Bacon, P., Morgan, J. F., Li, H., Williams, D. N., & Sanquist, T. (2018). *Human Factors Design Guidance for Level 2 and Level 3 Automated Driving Concepts (Report No. DOT HS 812 555)*. National Highway Traffic Safety Administration. https://www.nhtsa.gov/sites/nhtsa.gov/files/documents/13494_812555_1213automationhfguidance.pdf
- Chauvin, C., Said, F., & Langlois, S. (2023). Augmented reality HUD vs. Conventional HUD to perform a navigation task in a complex driving situation. *Cogn. Tech. Work.*, 25(2–3), 217–232. <https://doi.org/10.1007/s10111-023-00725-7>
- Chen, W., Song, J., Wang, Y., Wu, C., Ma, S., Wang, D., Yang, Z., & Li, H. (2023). Inattention blindness to unexpected hazard in augmented reality head-up display assisted driving: The impact of the relative position between stimulus and augmented graph. *Traffic Inj. Prev.*, 24(4), 344–351. <https://doi.org/10.1080/15389588.2023.2186735>
- Cheng, Y., Zhong, X., & Tian, L. (2023). Does the AR-HUD system affect driving behaviour? An eye-tracking experiment study. *Transportation Research Interdisciplinary Perspectives*, 18, Article 100767. <https://doi.org/10.1016/j.trip.2023.100767>
- Choudhary, P., & Velaga, N. R. (2019). Gap acceptance behavior at unsignalized intersections: Effects of using a phone and a music player while driving. *Traffic Inj. Prev.*, 20(4), 372–377. <https://doi.org/10.1080/15389588.2019.1591619>
- Cvahte Ojsteršek, T., & Topolšek, D. (2019). Influence of drivers' visual and cognitive attention on their perception of changes in the traffic environment. *Eur. Transp. Res. Rev.*, 11(1), 45. <https://doi.org/10.1186/s12544-019-0384-2>
- de Oliveira Faria, N. (2020). Evaluating Automotive Augmented Reality Head-up Display Effects on Driver Performance and Distraction. *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 553–554. 10.1109/VRW50115.2020.00128.
- de Oliveira Faria, N., & Gabbard, J. L. (2020). How long can a driver (Safely) glance at an augmented-reality head-up display? *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 64(1), 42–46. <https://doi.org/10.1177/1071181320641014>
- de Oliveira Faria, N., Gabbard, J. L., & Smith, M. (2020). Place in the World or Place on the Screen? Investigating the Effects of Augmented Reality Head-Up Display User Interfaces on Drivers' Spatial Knowledge Acquisition and Glance Behavior. *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 762–763. 10.1109/VRW50115.2020.00232.
- Dou, J., Xu, C., Chen, S., Xue, C., & Li, X. (2022). AR HUD interface optimization model for balancing driver's visual sensitivity and fatigue. *Procedia Comput. Sci.*, 214, 1568–1580. <https://doi.org/10.1016/j.procs.2022.11.345>
- Eyraud, R., Zibetti, E., & Baccino, T. (2015). Allocation of visual attention while driving with simulated augmented reality. *Transport. Res. F: Traffic Psychol. Behav.*, 32, 46–55. <https://doi.org/10.1016/j.trf.2015.04.011>
- Feierle, A., Beller, D., & Bengler, K. (2019). Head-up displays in urban partially automated driving: Effects of using augmented reality. *IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019, 1877–1882. <https://doi.org/10.1109/ITSC.2019.8917472>
- Feierle, A., Schlichtherle, F., & Bengler, K. (2022). Augmented reality head-up display: A visual support during malfunctions in partially automated driving? *IEEE Trans. Intell. Transp. Syst.*, 23(5), 4853–4865. <https://doi.org/10.1109/TITS.2021.3119774>
- Firth, M. (2019). *Introduction to automotive augmented reality head-up displays* (p. 10). Texas Instruments.
- Gabbard, J. L., Mehra, D. G., & Swan, J. E. (2019). Effects of AR display context switching and focal distance switching on human performance. *IEEE Trans. Vis. Comput. Graph.*, 25(6), 2228–2241. <https://doi.org/10.1109/TVCG.2018.2832633>
- Gabbard, J. L., Smith, M., Tanous, K., Kim, H., & Jonas, B. (2019). AR DRIVESIM: An immersive driving simulator for augmented reality head-up display research. *Frontiers in Robotics and AI*, 6(Journal Article), 98. <https://doi.org/10.3389/frobt.2019.00098>
- Gabbard, J. L., Smith, M., Merenda, C., Burnett, G., & Large, D. R. (2022). A perceptual color-matching method for examining color blending in augmented reality head-up display graphics. *IEEE Trans. Vis. Comput. Graph.*, 28(8), 2834–2851. <https://doi.org/10.1109/TVCG.2020.3044715>
- Gao, X., Werner, J., Necker, M., & Stork, W. (2019). A calibration method for automotive augmented reality head-up displays based on a consumer-grade mono-camera. *IEEE International Conference on Image Processing (ICIP)*, 2019, 4355–4359. <https://doi.org/10.1109/ICIP.2019.8803608>
- Genç, Ç., Soomro, S., Duyan, Y., Ölçer, S., Balci, F., Ürey, H., & Özcan, O. (2016). Head Mounted Projection Display & Visual Attention: Visual Attentional Processing of Head Referenced Static and Dynamic Displays while in Motion and Standing. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 1538–1547. 10.1145/2858036.2858449.
- Greatbatch, R. L., Kim, H., & Doerzaph, Z. (2022). The Effects of Augmented Reality Head-Up Display Graphics on Driver Situation Awareness and Takeover Performance in Driving Automation Systems. *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 521–522. 10.1109/VRW55335.2022.00116.
- Halit, L., Kemeny, A., Mohellebi, H., Garbay, S., Mérianne, F., Michelin, S., & Albaut, V. (2021). Augmented reality head-up-display for advanced driver assistance system: A driving simulation study. In *International Conference on Applied Human Factors and Ergonomics (AHFE)*. <https://doi.org/10.54941/ahfe100677>
- Hwang, Y., Park, B.-J., & Kim, K.-H. (2015). The effects of augmented-reality head-up display system on the perception of precautionary situation. *International Conference on Information and Communication Technology Convergence (ICTC)*, 2015, 1146–1148. <https://doi.org/10.1109/ICTC.2015.7354760>
- Hwang, Y., Park, B.-J., & Kim, K.-H. (2016). The effects of augmented-reality head-up display system usage on drivers? Risk Perception and Psychological Change. *ETRI J.* <https://doi.org/10.4218/etrij.16.0115.0770>
- International Organization for Standardization. (2017). *Road vehicles—Ergonomic aspects of transport information and control systems—Specifications and test procedures for in-vehicle visual presentation (ISO 15008:2017)*. International Organization for Standardization. <https://www.iso.org/standard/62784.html>
- Kalinina, A., & Putilin, A. (2020). Wide-field-of-view augmented reality eyeglasses using curved wedge waveguide. *Digital Optics for Immersive Displays II*, 11350B, 27–34. <https://doi.org/10.1117/12.2559320>
- Kim, H., & Gabbard, J. L. (2019). Assessing distraction potential of augmented reality head-up displays for vehicle drivers. *Human Factors: The Journal of the Human Factors and Ergonomics Society*. <https://doi.org/10.1177/0018720819844845>, 0018720819844844.

- Kim, H., Isleib, J. D., & Gabbard, J. L. (2016). Virtual shadow: Making cross traffic dynamics visible through augmented reality head up display. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 60(1), 2093–2097. <https://doi.org/10.1177/1541931213601474>
- Kim, H., Miranda Anon, A., Misu, T., Li, N., Tawari, A., & Fujimura, K. (2016). Look at me: Augmented reality pedestrian warning system using an in-vehicle volumetric head up display. In *Proceedings of the 21st International Conference on Intelligent User Interfaces* (pp. 294–298). <https://doi.org/10.1145/2856767.2856815>
- Kim, H., Wu, X., Gabbard, J. L., & Polys, N. F. (2013). Exploring head-up augmented reality interfaces for crash warning systems. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (pp. 224–227). <https://doi.org/10.1145/2516540.2516566>
- Lagoo, R., Charissis, V., & Harrison, D. K. (2019). Mitigating driver's distraction: Automotive head-up display and gesture recognition system. *IEEE Consum. Electron. Mag.*, 8(5), 79–85. <https://doi.org/10.1109/MCE.2019.2923896>
- Lauber, F., & Butz, A. (2013). Are HMDs the better HUDs? *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2013, 267–268. <https://doi.org/10.1109/ISMAR.2013.6671798>
- Lee, J., Koo, J., Park, J., & Lee, M.-C. (2021). Vehicle augmented reality head-up display information visualization enhancement algorithm and system. *International Conference on Information and Communication Technology Convergence (ICTC)*, 2021, 701–706. <https://doi.org/10.1109/ICTC52510.2021.9620873>
- Lee, J., Yanusik, I., Choi, Y., Kang, B., Hwang, C., Park, J., Nam, D., & Hong, S. (2020). Automotive augmented reality 3D head-up display based on light-field rendering with eye-tracking. *Opt. Express*, 28(20), 29788. <https://doi.org/10.1364/OE.404318>
- Lee, S. C., Hwangbo, H., & Ji, Y. G. (2016). Perceived visual complexity of in-vehicle information display and its effects on glance behavior and preferences. *International Journal of Human-Computer Interaction*, 32(8), 654–664. <https://doi.org/10.1080/10447318.2016.1184546>
- Lee, S. C., Kim, Y. W., & Ji, Y. G. (2019). Effects of visual complexity of in-vehicle information display: Age-related differences in visual search task in the driving context. *Appl. Ergon.*, 81, Article 102888. <https://doi.org/10.1016/j.apergo.2019.102888>
- Lee, Y.-H., Zhan, T., & Wu, S.-T. (2019). Prospects and challenges in augmented reality displays. *Virtual Reality & Intelligent Hardware*, 1(1), 10–20. <https://doi.org/10.3724/SP.J.2096-5796.2018.0009>
- Li, G., Wang, Y., Zhu, F., Sui, X., Wang, N., Qu, X., & Green, P. (2019). Drivers' visual scanning behavior at signalized and unsignalized intersections: A naturalistic driving study in China. *J. Saf. Res.*, 71, 219–229. <https://doi.org/10.1016/j.jsr.2019.09.012>
- Li, L., Zhong, B., Huttmacher, C., Liang, Y., Horrey, W. J., & Xu, X. (2020). Detection of driver manual distraction via image-based hand and ear recognition. *Accid. Anal. Prev.*, 137, Article 105432. <https://doi.org/10.1016/j.aap.2020.105432>
- Li, X., Rong, J., Li, Z., Zhao, X., & Zhang, Y. (2022). Modeling drivers' acceptance of augmented reality head-up display in connected environment. *Displays*, 75, Article 102307. <https://doi.org/10.1016/j.displa.2022.102307>
- Liang, Y., Zheng, P., & Xia, L. (2023). A visual reasoning-based approach for driving experience improvement in the AR-assisted head-up displays. *Adv. Eng. Inf.*, 55, Article 101888. <https://doi.org/10.1016/j.aei.2023.101888>
- Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C., Ioannidis, J. P., Clarke, M., Devereaux, P. J., Kleijnen, J., & Moher, D. (2009). The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: Explanation and elaboration. *Ann. Internal Med.*, 151. <https://doi.org/10.7326/0003-4819-151-4-200908180-00136>. W-65.
- Lisle, L., Merenda, C., Tanous, K., Kim, H., Gabbard, J. L., & Bowman, D. A. (2019). Effects of volumetric augmented reality displays on human depth judgments: implications for heads-up displays in transportation. *International Journal of Mobile Human Computer Interaction*, 11(2), 1–18. <https://doi.org/10.4018/IJMHCI.2019040101>
- Liu, W., Zhu, Y., Huang, R., Ohashi, T., Auernhammer, J., Zhang, X., Shi, C., & Wang, L. (2023). Designing interactive glazing through an engineering psychology approach: Six augmented reality scenarios that envision future car human-machine interface. *Virtual Reality & Intelligent Hardware*, 5(2), 157–170. <https://doi.org/10.1016/j.vrih.2022.07.004>
- Lv, Z., Liu, J., & Xu, L. (2021). A multi-plane augmented reality head-up display system based on volume holographic optical elements with large area. *IEEE Photonics J.*, 13(5), 1–8. <https://doi.org/10.1109/JPHOT.2021.3105670>
- Ma, X., Jia, M., Hong, Z., Kwok, A. P. K., & Yan, M. (2021). Does augmented-reality head-up display help? a preliminary study on driving performance through a VR-simulated eye movement analysis. *IEEE Access*, 9, 129951–129964. <https://doi.org/10.1109/ACCESS.2021.3112240>
- Maag, C., Schömg, N., Naujoks, F., Karl, I., Keinath, A., & Neukum, A. (2023). Measuring workload effects of augmented reality head-up displays using detection response task. *Transport. Res. F: Traffic Psychol. Behav.*, 92, 201–219. <https://doi.org/10.1016/j.trf.2022.11.010>
- Merenda, C., Kim, H., Tanous, K., Gabbard, J. L., Feichtl, B., Misu, T., & Suga, C. (2018). Augmented reality interface design approaches for goal-directed and stimulus-driven driving tasks. *IEEE Transactions on Visualization and Computer Graphics*, 24(11), 2875–2885. <https://doi.org/10.1109/TVCG.2018.2868531>
- Merenda, C., Suga, C., Gabbard, J. L., & Misu, T. (2019). Effects of “real-world” visual fidelity on AR interface assessment: A case study using AR head-up display graphics in driving. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2019, 145–156. <https://doi.org/10.1109/ISMAR.2019.00-10>
- Mukhopadhyay, A., Sharma, V. K., Tatyrao, P. G., Shah, A. K., Rao, A. M. C., Subin, P. R., & Biswas, P. (2023). A comparison study between XR interfaces for driver assistance in take over request. *Transportation Engineering*, 11, Article 100159. <https://doi.org/10.1016/j.treng.2022.100159>
- Murugan, S., Sampathkumar, A., Kanaga Suba Raja, S., Ramesh, S., Manikandan, R., & Gupta, D. (2022). Autonomous vehicle assisted by heads up display (HUD) with augmented reality based on machine learning techniques. In A. E. Hassanien, D. Gupta, A. Khanna, & A. Slowik (Eds.), *Virtual and Augmented Reality for Automobile Industry: Innovation Vision and Applications* (pp. 45–64). Springer International Publishing. https://doi.org/10.1007/978-3-030-94102-4_3
- Naujoks, F., Grattenthaler, H., Neukum, A., Weidl, G., & Petrich, D. (2015). Effectiveness of advisory warnings based on cooperative perception. *IET Intel. Transport Syst.*, 9(6), 606–617. <https://doi.org/10.1049/iet-its.2014.0190>
- Park, J., & Park, W. (2019). Functional requirements of automotive head-up displays: A systematic review of literature from 1994 to present. *Appl. Ergon.*, 76, 130–146. <https://doi.org/10.1016/j.apergo.2018.12.017>
- Pathivada, B. K., & Perumal, V. (2019). Analyzing dilemma driver behavior at signalized intersection under mixed traffic conditions. *Transport. Res. F: Traffic Psychol. Behav.*, 60, 111–120. <https://doi.org/10.1016/j.trf.2018.10.010>
- Pullukat, J., Tanaka, S., & Jiang, J. (2020). P-25: Effects of image distance on cognitive tunneling with augmented reality head up displays. *SID Symposium Digest of Technical Papers*, 51(1), 1427–1430. <https://doi.org/10.1002/sdtp.14155>
- Ren, D., Goldschwendt, T., Chang, Y., & Hollerer, T. (2016). Evaluating wide-field-of-view augmented reality with mixed reality simulation. *IEEE Virtual Reality (VR)*, 2016, 93–102. <https://doi.org/10.1109/VR.2016.7504692>
- Richter, P., von Spiegel, W., & Waldern, J. (2018). 55–2: *Invited paper*: Volume optimized and mirror-less holographic waveguide augmented reality head-up display. *SID Symposium Digest of Technical Papers*, 49(1), 725–728. <https://doi.org/10.1002/sdtp.12382>
- SAE. (2012). *Development of design and engineering recommendations for in-vehicle alphanumeric messages (SAE J2831_201204)*. SAE Mobilus. https://www.sae.org/standards/content/j2831_201204/
- Sburlan, D. F., Bautu, E., Puchianu, C. M., & Popovici, D. M. (2020). Adaptive interactive displaying system for in-vehicle use. *Procedia Comput. Sci.*, 176, 195–204. <https://doi.org/10.1016/j.procs.2020.08.021>
- Schall, M. C., Rusch, M. L., Lee, J. D., & Dawson, J. D. (2014). *Augmented Reality Cues and Elderly Driver Hazard Perception*.
- Schwarz, F., & Fastenmeier, W. (2017). Augmented reality warnings in vehicles: Effects of modality and specificity on effectiveness. *Accid. Anal. Prev.*, 101, 55–66. <https://doi.org/10.1016/j.aap.2017.01.019>
- Schwarz, F., & Fastenmeier, W. (2018). Visual advisory warnings about hidden dangers: Effects of specific symbols and spatial referencing on necessary and unnecessary warnings. *Appl. Ergon.*, 72(Journal Article), 25–36. <https://doi.org/10.1016/j.apergo.2018.05.001>
- Stoll, C. R. T., Izadi, S., Fowler, S., Green, P., Suls, J., & Colditz, G. A. (2019). The value of a second reviewer for study selection in systematic reviews. *Res. Synth. Methods*, 10(4), 539–545. <https://doi.org/10.1002/jrsm.1369>
- Strle, G., Košir, A., Sodnik, J., & Pečecnik, K. S. (2023). Physiological signals as predictors of cognitive load induced by the type of automotive head-up display. *IEEE Access*, 11, 87835–87848. <https://doi.org/10.1109/ACCESS.2023.3305383>

- Tabone, W., Happee, R., García, J., Lee, Y. M., Lupetti, M. L., Merat, N., & De Winter, J. (2023). Augmented reality interfaces for pedestrian-vehicle interactions: An online study. *Transport. Res. F: Traffic Psychol. Behav.*, 94, 170–189. <https://doi.org/10.1016/j.trf.2023.02.005>
- Instruments, T. (2021). The importance of a longer VID for AR HUDs. *Texas Instruments*. https://www.ti.com/lit/wp/dlpa117a/dlpa117a.pdf?ts=1679291061375&ref_url=https%253A%252F%252Fwww.google.com%252F
- Topliss, B. H., Harvey, C., & Burnett, G. (2020). How Long Can a Driver Look? Exploring Time Thresholds to Evaluate Head-up Display Imagery. *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 9–18. 10.1145/3409120.3410663.
- Topliss, B. H., Pampel, S. M., Burnett, G., Skrypchuk, L., & Hare, C. (2019). Follow the leader: Examining real and augmented reality lead vehicles as driving navigational aids. *International Journal of Mobile Human Computer Interaction*, 11(2), 19–38. <https://doi.org/10.4018/IJMHCI.2019040102>
- Valimont, R. B., Gangadharan, S., Vincenzi, D., & Majoros, A. (2007). The effectiveness of augmented reality as a facilitator of information acquisition in aviation maintenance applications. *Journal of Aviation/Aerospace Education & Research*. 10.15394/jaaer.2007.1478.
- Wang, W., Zhu, X., Chan, K., & Tsang, P. (2018). Digital Holographic System for Automotive Augmented Reality Head-Up-Display. *2018 IEEE 27th International Symposium on Industrial Electronics (ISIE)*, 1327–1330. 10.1109/ISIE.2018.8433601.
- Xu, N., Yu, F., Xu, J., Jin, D., Feng, H., Xu, Z., Li, Q., & Chen, Y. (2023). HUDNet: A dynamic calibration method for automotive augmented reality head-up-displays. *Displays*, 78, Article 102453. <https://doi.org/10.1016/j.displa.2023.102453>
- Yamin, P. A. R., Park, J., & Kim, H. K. (2021). Towards a Human-Machine Interface Guidance for In-Vehicle Augmented Reality Head-Up Displays. 10.24507/iceicel.15.12.1313.
- Yamin, P. A. R., Park, J., Kim, H. K., & Hussain, M. (2023). Effects of button colour and background on augmented reality interfaces. *Behav. Inform. Technol.*, 1–14. <https://doi.org/10.1080/0144929X.2023.2183060>
- Yontem, A. O., Li, K., Chu, D., Meijering, V., & Skrypchuk, L. (2021). Prospective immersive human-machine interface for future vehicles: Multiple zones turn the full windscreen into a head-up display. *IEEE Veh. Technol. Mag.*, 16(1), 83–92. <https://doi.org/10.1109/MVT.2020.3013832>
- Yoon, S. H., Lim, J., & Ji, Y. G. (2015). Assessment model for perceived visual complexity of automotive instrument cluster. *Appl. Ergon.*, 46, 76–83. <https://doi.org/10.1016/j.apergo.2014.07.005>
- Zhan, T., Yin, K., Xiong, J., He, Z., & Wu, S.-T. (2020). Augmented reality and virtual reality displays: Perspectives and challenges. *IScience*, 23(8), Article 101397. <https://doi.org/10.1016/j.isci.2020.101397>
- Zhang, Y., Yang, T., Zhang, X., Zhang, Y., & Sun, Y. (2021). Effects of full windshield head-up display on visual attention allocation. *Ergonomics*, 64(10), 1310–1321. <https://doi.org/10.1080/00140139.2021.1912398>
- Zhong, X., Cheng, Y., & Tian, L. (2022). Color Visibility Evaluation of In-Vehicle AR-HUD Under Different Illuminance. *Proceedings of the International Conference on Information Economy, Data Modeling and Cloud Computing, ICIDC 2022, 17-19 June 2022, Qingdao, China*. Proceedings of the International Conference on Information Economy, Data Modeling and Cloud Computing, ICIDC 2022, 17-19 June 2022, Qingdao, China, Qingdao, People's Republic of China. 10.4108/eai.17-6-2022.2322686.
- Zhou, X., & Yin, G. (2021). Research on symbol color of automotive augmented reality head-up display. *J. Phys. Conf. Ser.*, 1875(1), Article 012013. <https://doi.org/10.1088/1742-6596/1875/1/012013>

Putra A.R. Yamin is a Ph.D. candidate at Incheon National University and an assistant professor at Bandung Institute of Technology. He earned his masters degree in Human Factors and Ergonomics from the University of Nottingham in 2015. His research interest is on human-vehicle interaction.

Jaehyun Park is an associate professor at the Department of Industrial and Management Engineering, Incheon National University. He received his Ph.D. in Industrial and Management Engineering from Pohang University of Science and Technology in 2013. His areas of research include user value, user experience, and human-computer interaction.

Hyun K. Kim is an associate professor at the School of Information Convergence, Kwangwoon University. She received her Ph.D. in Industrial and Management Engineering from Pohang University of Science and Technology in 2013. Her research interests include artificial intelligence, behavioral sciences computing, physiology, and regression analysis.